

Introduction to Algorithms, Fourth Edition

Authors: Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein

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Chapter 1: The Role of Algorithms in Computing

1.1 Algorithms

Informally, an algorithm is any well-defined computational procedure that takes some value, or set of values, as input and produces some value, or set of values, as output in a finite amount of time.

1.2 Algorithms as a technology

If computers were infinitely fast and computer memory was free, would you have any reason to study algorithms? The answer is yes.